Structured notes: Abstract Algebra I, Module 2

Fraleigh, *A First Course in Abstract Algebra* (8th ed.), Module 2 (sections 5, 6 and 8 to 8.18). The syllabus names the 7th edition; items are matched by content. Numbers are the book's own. Formal version of the same content, with numbered results and the same order of items. Exercises: only the exercises the text names are reproduced, each directly after the item that names it. Figures are described with their labels; not redrawn. Item 8.19 onward is Module III and is not in these notes.

Section 5: Subgroups

Printed pp. 52–56 (5.1–5.26; Tables 5.8, 5.9; Figures 5.10, 5.27)

Prerequisites

- Group axioms and the notation of Section 2 (Definition 2.1).
- The determinant: $\det(AB) = \det(A)\det(B)$; A is invertible if and only if $\det(A) \neq 0$ (used in Example 5.14).
- Complex numbers of modulus 1 and their powers (Examples 5.5, 5.26).

1. Subgroups

Definition 5.1

DEFINITION 5.1 Let G be a group. A subset H of G that is closed under the binary operation of G, and that is a group under the induced operation, is a **subgroup of G**. Write $H \leq G$; write $H < G$ if $H \leq G$ and $H \neq G$.

Remark after Definition 5.1. $\langle \mathbb{Z}, + \rangle < \langle \mathbb{R}, + \rangle$. The set $\mathbb{Q}^+ \subset \mathbb{R}$, but $\langle \mathbb{Q}^+, \cdot \rangle \not\leq \langle \mathbb{R}, + \rangle$, because the operation differs.

Every group G has G and $\{e\}$ as subgroups.

DEFINITION 5.2 The subgroup G is the **improper subgroup** of G; every other subgroup is **proper**. The subgroup $\{e\}$ is the **trivial subgroup**; every other subgroup is **nontrivial**.

Examples 5.3 to 5.7

Example 5.3. In the additive group $\mathbb{R}^n$, the set of row vectors with first component 0 is a subgroup.

Example 5.4. $\langle \mathbb{Q}^+, \cdot \rangle$ is a proper subgroup of $\langle \mathbb{R}^+, \cdot \rangle$.

Example 5.5. $U_n \leq U \leq \mathbb{C}^*$, where U_n is the set of n th roots of unity, U is the set of complex numbers of absolute value 1, and $\mathbb{C}^*$ is the group of nonzero complex numbers under multiplication.

Example 5.6. $D_n \subseteq S_{\mathbb{Z}_n}$, and both are groups under composition; hence $D_n \leq S_{\mathbb{Z}_n}$.

Example 5.7. Up to isomorphism there are two groups of order 4 (see Exercise 20 of Section 2, reproduced right after this example). They are $\mathbb{Z}_4$ (Table 5.8) and the Klein 4-group V (Table 5.9). In $\mathbb{Z}_4$ the only nontrivial proper subgroup is $\{0, 2\}$; $\{0, 3\}$ is not a subgroup because it is not closed. In V the nontrivial proper subgroups are $\{e, a\}$, $\{e, b\}$, $\{e, c\}$; $\{e, a, b\}$ is not a subgroup because $ab = c$.

EXERCISE 20, SECTION 2 (NAMED IN EXAMPLE 5.7) This exercise shows that there are two nonisomorphic group structures on a set of 4 elements. Let the set be $\{e, a, b, c\}$, with e the identity for the operation. A group table must begin as in Table 2.29 (Module 1 excerpt, printed page 31). The square with the question mark cannot be a. It must be e, or an element different from e and a; without loss of generality take that element to be b. If the square is e, the table can be completed in two ways to give a group. Find these two tables. (You need not check associativity.) If the square is b, the table can be completed in one way. Find that table. (Again, you need not check associativity.) Of the three tables, two give isomorphic groups. Say which two, and give the one-to-one onto relabelling that is an isomorphism.

a. Are all groups of 4 elements commutative?

b. Find a way to relabel the four matrices $\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \right\}$ so that the matrix multiplication table is identical to one you constructed. This shows that the table defines an associative operation.

c. Show that for a particular value of n, the group elements of Exercise 14 can be relabelled so that their table is identical to one you constructed.

Table 5.8. $\mathbb{Z}_4$ under $+$

$+$	**0**	**1**	**2**	**3**
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

Table 5.9. The Klein 4-group V

V	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

Subgroup diagrams and two remarks

A subgroup diagram places the larger group higher, with a line from G down to H whenever $H \leq G$. **Figure 5.10.** (a) The subgroup diagram of $\mathbb{Z}_4$: $\mathbb{Z}_4 - \{0, 2\} - \{0\}$. (b) The subgroup diagram of V: V above the three subgroups $\{e, a\}$, $\{e, b\}$, $\{e, c\}$, above $\{e\}$.

Let $H \leq G$ and $a \in H$. The equation $ax = a$ has a unique solution in H, namely the identity of H; it has a unique solution in G, namely the identity e of G. Hence the identity of H is e. Applying the same argument to $ax = e$ shows that the inverse of a in H is the inverse of a in G.

Subgroup tests

Example 5.11. The continuous real-valued functions form a subgroup of the additive group F of all real-valued functions on $\mathbb{R}$.

> **THEOREM 5.12** A subset H of a group G is a subgroup if and only if (1) H is closed under the operation of G, (2) $e \in H$, and (3) $a \in H$ implies $a^{-1} \in H$.
>
> *Proof.* Necessity follows from Definition 5.1. For sufficiency, (2) and (3) give $\mathcal{G}_2$ and $\mathcal{G}_3$ in H, and $\mathcal{G}_1$ holds in H because it holds in G. ◆

Example 5.13. The differentiable functions form a subgroup of F, by Theorem 5.12.

Example 5.14. Let G be the multiplicative group of invertible $n \times n$ complex matrices and $T = \{A \in G \mid \det(A) = 1\}$. The identity $\det(AB) = \det(A)\det(B)$ shows that T is closed; $\det(I_n) = 1$; and $\det(A) = 1$ gives $\det(A^{-1}) = 1$. Hence $T \leq G$ by Theorem 5.12.

> **PLUS TWO RECALL: THE DETERMINANT OF A PRODUCT (STATUS OPEN, CHECK AGAINST THE KERALA TEXT)**
>
> **Recall.** $\det(AB) = \det(A)\det(B)$; A is invertible if and only if $\det(A) \neq 0$.
>
> **Re-derive.**
>
> 1. Put $B = A^{-1}$: $\det(A)\det(A^{-1}) = \det(I_n) = 1$.
> 2. If $\det(A) = 1$, then $\det(A^{-1}) = 1$.
>
> **Saved.** Example 5.14.

Theorem 5.15. A nonempty subset H of a group G is a subgroup of G if and only if, for all $a, b \in G$, $ab^{-1} \in G$. [as printed: the book writes G in both places. The intended statement is: for all $a, b \in H$, $ab^{-1} \in H$.] Proof: Exercise 51.

> **EXERCISE 51, SECTION 5 (THE PROOF OF THEOREM 5.15)** Prove Theorem 5.15.

Theorem 5.16. Let H be a finite nonempty subset of a group G. Then $H \leq G$ if and only if H is closed under the operation of G. Proof: Exercise 57.

> **EXERCISE 57, SECTION 5 (THE PROOF OF THEOREM 5.16)** Prove Theorem 5.16.

Example 5.17. By Theorem 5.16, $U_n \leq \mathbb{C}^*$: U_n has n elements, and $(z_1 z_2)^n = 1$ for $z_1, z_2 \in U_n$.

Example 5.18. $H = \{\iota, \rho, \ldots, \rho^{n-1}\} \leq D_n$, since $\rho^k \rho^r = \rho^{k+_n r} \in H$ and Theorem 5.16 applies.

2. Cyclic subgroups

In $\mathbb{Z}_{12}$, any subgroup containing 3 contains 3, 6, 9, 0; the set $\{0, 3, 6, 9\}$ is a subgroup and is the smallest one containing 3.

In a group G, a subgroup containing a contains a^n for every $n \in \mathbb{Z}$: positive powers by closure, $a^0 = e$, and negative powers as inverses. Powers need not be distinct: in V, $a^2 = e$, $a^3 = a$, $a^4 = e$, $a^{-1} = a$.

> **THEOREM 5.19** Let $a \in G$. Then $H = \{a^n \mid n \in \mathbb{Z}\}$ is a subgroup of G, and it is the smallest subgroup of G containing a: every subgroup containing a contains H.
>
> *Proof.* $a^r a^s = a^{r+s}$, $a^0 = e$, and $(a^r)^{-1} = a^{-r} \in H$; by Theorem 5.12, $H \leq G$. Any subgroup containing a contains H by the remark above. ◆

Footnote on the book's terms: a subset is *minimal* with a property if no proper subset has it; it is the *smallest* if it is contained in every subset with the property. $\{e, a\}$, $\{e, b\}$, $\{e, c\}$ are minimal

nontrivial subgroups of V; V has no smallest nontrivial subgroup.

DEFINITION 5.20 For $a \in G$, the subgroup $\{a^n \mid n \in \mathbb{Z}\}$ of Theorem 5.19 is the **cyclic subgroup generated by** a, denoted $\langle a \rangle$.

Example 5.21. In D_{10}: $\langle \mu\rho^k \rangle = \{\iota, \mu\rho^k\}$, since $(\mu\rho^k)^2 = \iota$; and $\langle \rho \rangle = \{\iota, \rho, \ldots, \rho^9\}$, since $\rho^{-1} = \rho^9$ and $\rho^{10} = \iota$.

DEFINITION 5.22 a **generates** G, and is a **generator for** G, if $\langle a \rangle = G$. G is **cyclic** if it has a generator.

Example 5.23. $\mathbb{Z}_4$ is cyclic with generators 1 and 3. V is not cyclic: $\langle a \rangle$, $\langle b \rangle$, $\langle c \rangle$ are proper of order 2.

Example 5.24. $\mathbb{Z}$ is cyclic with generators 1 and -1 only. $\mathbb{Z}_n$ is cyclic; for $n > 1$, 1 and $n-1$ are generators.

Example 5.25. $\langle 3 \rangle = 3\mathbb{Z}$ in $\mathbb{Z}$; in general $n\mathbb{Z} = \langle n \rangle$, and $6\mathbb{Z} < 3\mathbb{Z}$.

Example 5.26. $U_n = \langle \zeta \rangle$ with $\zeta = \cos\frac{2\pi}{n} + i\sin\frac{2\pi}{n}$. Powers of ζ move counterclockwise and reach each element of U_n in turn (Figure 5.27). U_n is cyclic, and ζ is a generator.

Figure 5.27. The points $1, \zeta, \zeta^2, \zeta^3$ and so on, equally spaced on the unit circle, with i marked. Labels as printed; not redrawn.

Section 6: Cyclic Groups

Printed pp. 61–67 (6.1–6.21; Figures 6.3, 6.11, 6.12, 6.19)

Prerequisites

- Cyclic subgroups $\langle a \rangle$, generators, Theorem 5.19 and Definitions 5.20, 5.22.
- Integer division with remainder (the division algorithm, Theorem 6.2).
- Isomorphism of groups (Section 2).

1. Elementary properties

THEOREM 6.1 Every cyclic group is abelian.

Proof. Let $G = \langle a \rangle$ and $g_1 = a^r$, $g_2 = a^s$. Then $g_1 g_2 = a^{r+s} = a^{s+r} = g_2 g_1$. ◆

Multiplicative notation is used throughout this section, whether or not the group is abelian.

2. The division algorithm and the gcd

THEOREM 6.2 (DIVISION ALGORITHM FOR $\mathbb{Z}$) If $m \in \mathbb{Z}^+$ and $n \in \mathbb{Z}$, there exist unique $q, r \in \mathbb{Z}$ with $n = mq + r$ and $0 \le r < m$.

Proof. The book argues from the number line (Figure 6.3): either $n = qm$, or n lies strictly between qm and $(q+1)m$ for a unique q, and $r = n - qm$. ◆

Figure 6.3. Number-line diagrams for the cases $n \ge 0, q \ge 0$ and $n < 0, q < 0$, with the remainder r marked. Labels as printed; not redrawn.

Example 6.4. $38 = 7(5) + 3$: quotient 5, remainder 3.

Example 6.5. $-38 = 7(-6) + 4$: quotient -6, remainder 4.

THEOREM 6.6 A subgroup of a cyclic group is cyclic.

Proof. Let $G = \langle a \rangle$ and $H \le G$. If $H = \{e\}$, then $H = \langle e \rangle$. Otherwise let $m \in \mathbb{Z}^+$ be least with $a^m \in H$, and put $c = a^m$. For $b = a^n \in H$, write $n = mq + r$, $0 \le r < m$. Then $a^r = c^{-q} b \in H$, so $r = 0$ by minimality, and $b = c^q$. Thus $H = \langle c \rangle$. ◆

COROLLARY 6.7 The subgroups of $\langle \mathbb{Z}, + \rangle$ are precisely the groups $n\mathbb{Z}$, $n \in \mathbb{Z}$.

EXERCISE 54, SECTION 6 (NAMED IN THE REMARK AFTER COROLLARY 6.7) Let r and s be integers. Show that $\{nr + ms \mid n, m \in \mathbb{Z}\}$ is a subgroup of $\mathbb{Z}$.

DEFINITION 6.8 Let $r \in \mathbb{Z}^+$, $s \in \mathbb{Z}_{\ge 0}$. The positive generator d of $H = \{nr + ms \mid n, m \in \mathbb{Z}\}$ is the **greatest common divisor** $\gcd(r, s)$.

Since $r, s \in d\mathbb{Z}$, d is a common divisor; since $d = nr + ms$, every common divisor of r and s divides d. Hence d is the greatest common divisor. The identity $d = nr + ms$ is **Bézout's identity**.

Example 6.9. $\gcd(42, 72) = 6 = (3)(72) + (-5)(42)$. Integers r, s are **relatively prime** if $\gcd(r, s) = 1$.

$$\text{If } \gcd(r, s) = 1 \text{ and } r \mid sm, \text{ then } r \mid m. \qquad (1)$$

Proof of (1). $1 = ar + bs$ gives $m = arm + bsm$; r divides both terms. ◆

3. Structure of cyclic groups

THEOREM 6.10 Let $G = \langle a \rangle$. If $|G| = \infty$, then $G \cong \langle \mathbb{Z}, + \rangle$. If $|G| = n$, then $G \cong \langle \mathbb{Z}_n, +_n \rangle$.

Proof. **Case I**: $a^m \ne e$ for all $m > 0$. Then $a^h = a^k$ forces $h = k$, and $\phi(a^i) = i$ is a well-defined bijection $G \to \mathbb{Z}$ with $\phi(a^i a^j) = \phi(a^i) + \phi(a^j)$. **Case II**: let n be least positive with $a^n = e$. Then $a^0, \ldots, a^{n-1}$ are distinct and exhaust G; $\psi(a^i) = i$ is a bijection $G \to \mathbb{Z}_n$ with $\psi(a^i a^j) = \psi(a^i) +_n \psi(a^j)$. ◆

Figures 6.11 and 6.12. The elements of a cyclic group of order n placed evenly on a circle; labels are $a^0, a, \ldots, a^{n-1}$ in 6.11 and $0, 1, \ldots, n-1$ in 6.12. Not redrawn.

Example 6.13. Multiplication is addition of positions around the circle: $a^h a^k = a^r$, where $h + k = nq + r$ and $0 \le r < n$. This is the isomorphism $\langle a \rangle \cong \mathbb{Z}_n$.

Consequently, the order of $a \in G$ is the least $n \in \mathbb{Z}^+$ with $a^n = e$.

Example 6.14. The order of a k-cycle is k.

4. Subgroups of finite cyclic groups

THEOREM 6.15 Let G be cyclic of order n with generator a, and let $b = a^s$. Then b generates a cyclic subgroup of order n/d, where $d = \gcd(n, s)$. Moreover $\langle a^s \rangle = \langle a^t \rangle$ if and only if $\gcd(s, n) = \gcd(t, n)$.

Proof. $|\langle b \rangle|$ is the least $m > 0$ with $n \mid ms$. Write $d = un + vs$; then $\gcd(n/d, s/d) = 1$, and by (1) the least such m is n/d. The second statement follows from the uniqueness of the subgroup of each order n/d in $\mathbb{Z}_n$. ◆

Example 6.16. In $\mathbb{Z}_{12}$: $\langle 3 \rangle$ has 4 elements, $\langle 8 \rangle$ has 3, and 5 generates the whole group.

COROLLARY 6.17 If a generates a finite cyclic group of order n, then its generators are the a^r with $\gcd(r, n) = 1$.

Example 6.18. The subgroups of $\mathbb{Z}_{18}$ are $\langle 1 \rangle = \mathbb{Z}_{18}$, $\langle 2 \rangle$ (order 9), $\langle 3 \rangle$ (order 6), $\langle 6 \rangle = \{0, 6, 12\}$, $\langle 9 \rangle = \{0, 9\}$ and $\langle 0 \rangle$. Figure 6.19 gives the subgroup diagram: $\langle 1 \rangle$ above $\langle 2 \rangle$ and $\langle 3 \rangle$; $\langle 2 \rangle$ above $\langle 6 \rangle$; $\langle 3 \rangle$ above $\langle 6 \rangle$ and $\langle 9 \rangle$; $\langle 6 \rangle$ and $\langle 9 \rangle$ above $\langle 0 \rangle$. Labels as printed; not redrawn.

COROLLARY 6.20 If G is finite and cyclic and $H \le G$, then $|H|$ divides $|G|$.

Proof. By Theorem 6.6 and Theorem 6.15, $|H| = n/\gcd(n, s)$ for some s, which divides n. ◆

Example 6.21. The subgroups of $\mathbb{Z}_{28}$ have orders $1, 2, 4, 7, 14, 28$, and each order occurs.

Corollary 6.20 holds without cyclicity; Lagrange's Theorem (Section 10) gives this for all finite groups.

Section 8: Groups of Permutations (8.1–8.18)

Printed pp. 77–82 (8.1–8.18; Table 8.9; Historical note)

Prerequisites

- Isomorphisms of groups (Section 2).
- Maps: composition, one-to-one and onto maps, inverse maps.
- Permutations and cycle notation (Section 4, used without reproduction).

1. Homomorphisms

DEFINITION 8.1 A map $\phi : G \to G'$ of groups is a **homomorphism** if $\phi(ab) = \phi(a)\phi(b)$ for all $a, b \in G$.

Example 8.2. $\phi : \mathbb{R} \to U$, $\phi(x) = \cos(2\pi x) + i\sin(2\pi x) = e^{2\pi i x}$, is a homomorphism. It maps onto U and is not one-to-one, so it is not an isomorphism. Moreover $\phi(-x) = (\phi(x))^{-1}$.

Example 8.3. $\phi : U_{28} \to U_4$, $\phi(z) = z^7$, is a well-defined homomorphism: $z^{28} = 1$ gives $(z^7)^4 = 1$, and $(z_1 z_2)^7 = z_1^7 z_2^7$.

PLUS TWO RECALL: FUNCTIONS, IMAGES AND INVERSE IMAGES (STATUS OPEN, CHECK AGAINST THE KERALA TEXT)

Recall. For $\phi : X \to Y$, the image $\phi[A]$ of a subset A and the inverse image $\phi^{-1}[B]$ of a subset B may be nonempty even when ϕ is not onto.

Re-derive.

1. $\phi(x) = x^2$ on $\mathbb{R}$: $\phi^{-1}[\{4\}] = \{2, -2\}$.
2. $\phi[\{-1, 1\}] = \{1\}$.

Saved. Definition 8.4.

DEFINITION 8.4 For $\phi : X \to Y$, $A \subseteq X$, $B \subseteq Y$: $\phi[A] = \{\phi(a) \mid a \in A\}$ is the **image of A**; $\phi^{-1}[B] = \{a \in X \mid \phi(a) \in B\}$ is the **inverse image of B**.

THEOREM 8.5 Let $\phi : G \to G'$ be a homomorphism.

1. $\phi(e) = e'$.
2. $\phi(a^{-1}) = \phi(a)^{-1}$ for all $a \in G$.
3. $H \leq G$ implies $\phi[H] \leq G'$.
4. $K' \leq G'$ implies $\phi^{-1}[K'] \leq G$.

Proof. (1) $\phi(e) = \phi(e)\phi(e)$, so $\phi(e) = e'$. (2) $e' = \phi(a)\phi(a^{-1})$. (3) $\phi(a)\phi(b) = \phi(ab) \in \phi[H]$; (1) and (2) supply the identity and inverses. (4) If $a, b \in \phi^{-1}[K']$, then $\phi(ab) = \phi(a)\phi(b) \in K'$, and $\phi(a^{-1}) = \phi(a)^{-1} \in K'$; also $e' \in K'$. ◆

DEFINITION 8.6 For a homomorphism $\phi : G \to G'$, the subgroup $\mathbf{Ker}(\phi) = \phi^{-1}[\{e'\}] = \{x \in G \mid \phi(x) = e'\}$ is the **kernel** of ϕ.

The kernel is used in the book to define the alternating group. The case $H = G$ of Theorem 8.5 (3) gives that $\phi[G]$ is a subgroup of G', which is used in Cayley's Theorem.

Example 8.7. In Example 8.2, $\mathbf{Ker}(\phi) = \mathbb{Z}$. For $n \geq 1$, $\phi[\langle \frac{1}{n} \rangle] = U_n$.

Example 8.8. $\phi : \mathbb{Z}_n \to D_n$, $\phi(k) = \rho^k$, is a homomorphism (cases $a + b < n$ and $a + b \geq n$, using $\rho^n = \iota$), with image $\langle \rho \rangle$.

2. Cayley's Theorem

Several standard groups are groups of permutations: $\mathbb{Z}_n \cong \langle (1, 2, \ldots, n) \rangle \leq S_n$; D_n acts on $\mathbb{Z}_n$; $GL(n, \mathbb{R})$ acts on $\mathbb{R}^n$. Cayley's Theorem says that every group is isomorphic to a group of permutations. The proof uses the group table: the row of a defines a permutation of G. For D_3 (Table 8.9), the row for $\mu\rho$ gives $\begin{pmatrix} \iota & \rho & \rho^2 & \mu & \mu\rho & \mu\rho^2 \\ \mu\rho & \mu\rho^2 & \mu & \rho^2 & \iota & \rho \end{pmatrix}$.

Table 8.9. The group table of D_3, headed $\iota, \rho, \rho^2, \mu, \mu\rho, \mu\rho^2$:

D_3	ι	ρ	ρ^2	μ	$\mu\rho$	$\mu\rho^2$
ι	ι	ρ	ρ^2	μ	$\mu\rho$	$\mu\rho^2$
ρ	ρ	ρ^2	ι	$\mu\rho^2$	μ	$\mu\rho$
ρ^2	ρ^2	ι	ρ	$\mu\rho$	$\mu\rho^2$	μ
μ	μ	$\mu\rho$	$\mu\rho^2$	ι	ρ^2	ρ
$\mu\rho$	$\mu\rho$	$\mu\rho^2$	μ	ρ^2	ι	ρ
$\mu\rho^2$	$\mu\rho^2$	μ	$\mu\rho$	ρ	ρ^2	ι

Write $\lambda_x(g) = xg$. This is well defined for infinite groups as well as finite ones.

DEFINITION 8.10 The map $\phi : G \to S_G$, $\phi(x) = \lambda_x$, is the **left regular representation** of G.

Each λ_x is a permutation: it is one-to-one by cancellation ($xa = xb \Rightarrow a = b$), and onto since $\lambda_x(x^{-1}b) = b$.

THEOREM 8.11 (CAYLEY'S THEOREM) Every group is isomorphic to a group of permutations.

Proof. If $\lambda_a = \lambda_b$, then $ae = be$, so $a = b$: ϕ is one-to-one. $\lambda_{ab}(g) = (ab)g = a(bg) = (\lambda_a\lambda_b)(g)$, so $\phi(ab) = \phi(a)\phi(b)$. By Theorem 8.5 (3) with $H = G$, $\phi[G] \leq S_G$, and $\phi : G \to \phi[G]$ is an isomorphism. ◆

Example 8.12. The subgroup from the proof is not always the smallest. D_n is isomorphic to a subgroup of $S_{\mathbb{Z}_n}$, while the proof gives a subgroup of S_{D_n}. And $\mathbb{Z}_6 \cong \langle (1,2,3)(4,5)\rangle \leq S_5$.

The right regular representation is defined in Definition 8.13.

HISTORICAL NOTE Arthur Cayley (1821–1895) gave an abstract definition of a group in 1854. In 1878 he stated Theorem 8.11, with a proof based on the group table. Walther von Dyck's axiomatic definition of 1882 led to the development of abstract group theory. The book's full note is kept in the excerpt.

DEFINITION 8.13 With $\sigma_x(g) = gx$, the map $\tau : G \to S_G$, $\tau(x) = \sigma_{x^{-1}}$, is the **right regular representation** of G.

It can also be used to prove Cayley's Theorem. See Exercise 54, printed next.

EXERCISE 54, SECTION 8 (NAMED AFTER DEFINITION 8.13) Prove Cayley's Theorem using the right regular representation rather than the left regular representation.

3. Transpositions, orbits

DEFINITION 8.14 A cycle of length 2 is a **transposition**.

$$(a_1, a_2, \ldots, a_n) = (a_1, a_n)(a_1, a_{n-1}) \cdots (a_1, a_3)(a_1, a_2).$$

Hence every n-cycle is a product of $n - 1$ transpositions.

THEOREM 8.15 Every permutation of a finite set with at least two elements is a product of transpositions.

Example 8.16. $(1,6)(2,5,3) = (1,6)(2,3)(2,5)$.

Example 8.17. In S_n ($n \geq 2$), $\iota = (1,2)(1,2)$.

The representation of a permutation as a product of transpositions is not unique. The even or odd parity of the number of transpositions is studied in the book from 8.19 onward (Module III, not in these notes), which uses the orbits defined below.

For $\sigma \in S_A$ and $a \in A$, the **orbit** of a is $\{\sigma^k(a) \mid k \in \mathbb{Z}\}$. For $\sigma \in S_n$, the orbit of a is the set of entries of the cycle containing a in the disjoint-cycle form of σ.

Example 8.18. For $\sigma = (1,2,6)(3,5) \in S_6$: the orbits of 1, 2, 6 are $\{1,2,6\}$; the orbits of 3, 5 are $\{3,5\}$; the orbit of 4 is $\{4\}$, since $\sigma = (1,2,6)(3,5)(4)$ when 1-cycles are included.